

CS 381 Project 1 Sample input/output:

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SLIDING PUZZLE SEARCH PROGRAM
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1. Run BFS
2. Run IDS
3. Run BFS and IDS
4. Run DLS
5. Quit
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Enter your choice (1, 2, 3, or 4): 3
Enter the START state as a vector of vectors:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 14, 15, 0]]
Enter the GOAL state as a vector of vectors:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [14, 15, 13, 0]]

Running BFS and IDS...
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BFS RESULT
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Number of moves needed: 18
Sequence of moves : ['L', 'L', 'L', 'U', 'R', 'D', 'R', 'R', 'U', 'L', 'L', 'L', 'D', 'R', 'R', 'U', 'R', 'D']
Move string      : LLLURDRRULLDRRURD
Nodes removed   : 844765
Maximum queue size : 954399
CPU time        : 5.092832 seconds
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IDS RESULT
=====
Number of moves needed : 18
Sequence of moves : ['L', 'L', 'L', 'U', 'R', 'D', 'R', 'R', 'U', 'L', 'L', 'L', 'D', 'R', 'R', 'U', 'R', 'D']
Move string      : LLLURDRRULLDRRURD
Nodes removed   : 2083996
Maximum stack size : 27
CPU time        : 4.833533 seconds
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Enter your choice (1, 2, 3, or 4): 4

Enter the START state as a vector of vectors:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 14, 15, 0]]

Enter the GOAL state as a vector of vectors:
[[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [14, 15, 13, 0]]
Enter the depth of search for DLS:17

Running DLS ...
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DLS RESULT
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Number of moves needed : 0 (Solution not found)
Sequence of moves : []
Move string      : (empty)
Nodes removed   : 819531
Maximum stack size : 26
CPU time        : 1.931086 seconds
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